

Counting

Chapter 6

With Question/Answer Animations

The Basics of Counting

Section 6.1



Counting

- **Figuring out how many possible outcomes there are without listing them all.**
- Instead of writing every possibility (which can be huge), we use rules to calculate the total quickly.

Basic Counting Principles: The Product Rule

- The **Product Rule** says:
 - If a process happens in steps, you **multiply** the number of choices at each step.
- **In simple terms:**
 - Step 1 has **n_1 choices**
 - Step 2 has **n_2 choices**
 - Total outcomes = **$n_1 \times n_2$**
- If there are more steps, just keep multiplying.

Example:

- You choose a shirt (3 options)
 - Then choose pants (2 options)
- Total outfits = **$3 \times 2 = 6$ outfits**
Because for each shirt, you can pair any pants.

The Product Rule

- **Example 1: Bit strings of length 7**
- A **bit** is either: 0, 1
- So each position has **2 choices**.
- There are **7 positions**, so:
- $\text{Total} = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$
 $= 2^7 = 128$
- So, there are **128 possible bit strings**

Example 2: License plates
3 letters + 3 digits

Condition:

- Each letter: 26 choices (A–Z)
- Each digit: 10 choices (0–9)

So:

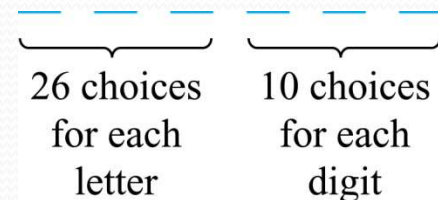
Total=

$$= 26 \times 26 \times 26 \times 10 \times 10 \times 10$$

$$= 26^3 \times 10^3$$

$$= 17,576,000$$

That's over 17 million possible plates!



Counting Functions

- A **function** means:
 - Every input (from the domain) chooses exactly **one output** (from the codomain)
- **Example:**
 - You have **m inputs**
 - Each input can go to **any of n outputs**
 - So for each input, you have **n choices**
 - Apply the product rule:
$$n \times n \times n \cdots (m \text{ times}) = n^m$$

Counting Functions

- 2. One-to-One Functions (Injective)
- Now there's a restriction:
 - No two inputs can go to the same output
 - So choices decrease each time.

Like:

- First input $\rightarrow$ n choices
- Second input $\rightarrow$ $n-1$ choices
- Third input $\rightarrow$ $n-2$ choices
- Keep going...

$$n \times (n-1) \times (n-2) \times \dots$$

This only works if $m \leq n$
(otherwise impossible to assign unique outputs)

$m = 3$ inputs

$n = 5$ outputs

Choices:

1st $\rightarrow$ 5

2nd $\rightarrow$ 4

3rd $\rightarrow$ 3

$$\text{Total} = 5 \times 4 \times 3 = 60$$



Telephone Numbering Plan

Example: The *North American numbering plan* (NANP) specifies that a telephone number consists of 10 digits, consisting of a three-digit area code, a three-digit office code, and a four-digit station code. There are some restrictions on the digits.

- Let X denote a digit from 0 through 9 $\rightarrow$ **10 choices**
- Let N denote a digit from 2 through 9 $\rightarrow$ **8 choices**
- Let Y denote a digit that is 0 or 1 $\rightarrow$ **2 choices**
- **In the old plan (in use in the 1960s) the format was $NYX-NNX-XXX$.**
- **In the new plan, the format is $NXX-NXX-XXX$.**

How many different telephone numbers are possible under the old plan and the new plan?

Telephone Numbering Plan...Continue

Area Code (NYX)

- $N \rightarrow 8$
- $Y \rightarrow 2$
- $X \rightarrow 10$
- $\text{Total} = 8 \times 2 \times 10 = 160$

Office Code (NNX)

- $N \rightarrow 8$
- $N \rightarrow 8$
- $X \rightarrow 10$

$$\text{Total} = 8 \times 8 \times 10 = 640$$

Station Code (XXXX)

- 4 digits $\rightarrow$ each 10 choices
- $\text{Total} = 10^4 = 10,000$

So,

$$160 \times 640 \times 10,000 = 1,024,000,000$$

Old plan: 1,024,000,000 numbers

Telephone Numbering Plan...Continue

Area Code (NXX)

- $N \rightarrow 8$
- $X \rightarrow 10$
- $X \rightarrow 10$

$$\text{Total} = 8 \times 10 \times 10 = 800$$

Office Code (NXX)= Same as above: 800

Station Code (XXXX)= 10,000

So,

$$800 \times 800 \times 10,000 = 6,400,000,000$$

New plan: 6,400,000,000 numbers

Product Rule in Terms of Sets

- If you have sets $A_1, A_2, \dots, A_m$,
- The Cartesian product means:

$$|A_1 \times A_2 \times \dots \times A_m|$$

- Apply product rule

$$|A_1| \cdot |A_2| \cdot \dots \cdot |A_m|.$$

- Let:
- $A = \{1, 2\}$ 2 $\rightarrow$ elements
- $B = \{x, y, z\}$ 3 $\rightarrow$ elements
- **Cartesian product:**
 - $A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}$
- Total = $2 \times 3 = 6$



Basic Counting Principles: The Sum Rule

If a task can be done in different **separate ways**,
just **add the number of ways**.

Different options → add them

If:

- one task can be done in n_1 ways
- another (separate) task in n_2 ways
- and they **don't overlap**
- **Total = $n_1 + n_2$**



Basic Counting Principles: The Sum Rule

Example: The mathematics department must choose either a student or a faculty member as a representative for a university committee. How many choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student.

Solution: By the sum rule it follows that there are $37 + 83 = 120$ possible ways to pick a representative.

Combining the Sum and Product Rule

- Use:
 - **Sum rule** ($+$) when you choose between cases (OR)
 - **Product rule** ($\times$) when steps happen together (AND)

Combining the Sum and Product Rule

Example: Suppose statement labels in a programming language can be either a single letter or a letter followed by a digit. Find the number of possible labels.

Solution:

- **Label is either:**
 - 1 letter OR
 - letter + digit
- **Step:**
 - Single letter: 26 ways
 - Letter + digit: $26 \times 10 = 260$
- **Total (sum rule): $26 + 260 = 286$**
- **Or direct: Use the product rule.**
 $26 + 26 \cdot 10 = 286$

Counting Passwords

- Combining the sum and product rule allows us to solve more complex problems.
Example: Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution: Let P be the total number of passwords, and let P_6 , P_7 , and P_8 be the passwords of length 6, 7, and 8.

- By the sum rule $P = P_6 + P_7 + P_8$.
- For each length:**
- Step 1: total possibilities (letters + digits)**
 36^n
- Step 2: remove invalid (only letters)**
 26^n
- So valid passwords:**
$$P_n = 36^n - 26^n$$
- So Final:**
 - $P = (36^6 - 26^6) + (36^7 - 26^7) + (36^8 - 26^8)$

Counting Passwords

- By the sum rule $P = P_6 + P_7 + P_8$.
- To find each of P_6 , P_7 , and P_8 , we find the number of passwords of the specified length composed of letters and digits and subtract the number composed only of letters. We find that:

$$P_6 = 36^6 - 26^6 = 2,176,782,336 - 308,915,776 \\ = 1,867,866,560.$$

$$P_7 = 36^7 - 26^7 = 78,364,164,096 - 8,031,810,176 \\ = 70,332,353,920.$$

$$P_8 = 36^8 - 26^8 = 2,821,109,907,456 - 208,827,064,576 \\ = 2,612,282,842,880.$$

Consequently, $P = P_6 + P_7 + P_8 = 2,684,483,063,360$.

Basic Counting Principles:

Subtraction Rule

- **Subtraction Rule:** When you **add two groups**, but some results are counted twice, you **subtract the overlap**.

- **Formula**

$$\text{Total} = n_1 + n_2 - (\text{common ways})$$

- Also known as, the *principle of inclusion-exclusion*:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

- **Example:**

- Ways from set A = 10

- Ways from set B = 8

- Overlap (common ways) = 3

- **So, $10+8-3=15$**

The Pigeonhole Principle

Section 6.2

The Pigeonhole Principle

- **The core idea**
 - If you have **more things than containers**, then **some container must hold more than one thing**.
- **Why this is true (no math first)**
- **Imagine:**
 - 3 pigeonholes (boxes)
 - 4 pigeons (objects)
- **Try to place them:**
 - Put 1 pigeon in each hole → 3 pigeons used
 - 1 pigeon is left
- That pigeon must go into an already filled hole
- So one hole has **at least 2 pigeons**
 - “You cannot evenly distribute too many things into too few places without overlap.” ◀



The Pigeonhole Principle

Corollary 1: A function f from a set with $k + 1$ elements to a set with k elements is not one-to-one.

Proof: Use the pigeonhole principle.

- Think of each output value in the codomain as a **box**.
- Think of each input value in the domain as an **object**.
- Place each input into the box of its output.

Since there are:

- **$k + 1$ inputs**
- **only k output boxes**
- At least one box must contain **two or more inputs**.

This means:

- Two different inputs have the same output.
- Therefore, the function **cannot be one-to-one**.



The Generalized Pigeonhole Principle

- Instead of just “2 in a box”, we ask:
- **how crowded must a box be?**

Formula idea

- If:
- N objects
- k boxes
- Then at least one box has:

$$\left\lceil \frac{N}{k} \right\rceil$$

“On average each box gets N/k items, so at least one box gets that much or more.”

• Example 1: Students in Classes

Suppose

- 50 students
- 6 classes

Calculate:

- Round up: $50/6 = 8.33$

- Answer: $\lceil 8.33 \rceil = 9$

- At least **one class** must have **9 students**.
- Because if every class had only 8 students:

$$6 \times 8 = 48$$

But there are 50 students, so 2 extra students must go somewhere.

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- **Example 2: Birth Months**
 - 100 people
 - 12 months
 - $100/12 = 8.33$
 - Round up: 9 Answer
 - At least **9 people** were born in the same month.